

Assignment 2 Write Up

Based on my observations from the data, this assignment and last assignment, I wanted to see if the number of EV sales per county was reflective of respective affluence and social status there. Due to the rise of Teslas, green energy, and overall reputation, EV's are now booming in popularity with companies like Waymo slowly reshaping the car landscape. Because of this, I sought to use my interactive visualization to see and reflect this hypothesis, that EV use is particularly popular amongst those in high-income, high-status counties. Based on our labs and activities in class, I wanted to create something that reflected what we had learned: a bar chart, Chloropleth-inspired map, and a line graph.

For the bar chart, I liked how it represented data in the first assignment, and I wanted to see how all the counties EV sales performed relative to one another since just the map visual would provide less numerical data. With the drop down, you can see how EV sales performed per county over time, with counties such as Los Angeles, Orange, and Santa Clara being the most high selling counties. Based on my knowledge about these counties both in affluence and status, I saw my hypothesis being slightly proven, however, I wanted to see how the data looked geographically.

I really liked the Chloropleth map we saw in class, and I wanted to create something similar to map the data. I found a free dataset for the counties and matched that to the county data (rather than the ZIP) to map the data for each county. I chose sequential coloring to represent concentration as I found it to be the most intuitive in our past activities, that a darker color represents "more" since there is more of that color present, and vice versa. This ended up displaying the bar chart data in a very visual way, displaying how the LA and SF areas once again had very high EV cars in those counties. It also made it very clear what areas are more rural and do not have many EV's.

And lastly, I wanted to see how the cars were sold over time respective to each county, to be able to see the booms as well as the slumps for EV sales. This was seen in the line chart where the dropdown for each county can be selected and the growth over time is displayed. The most interesting part about this visualization was seeing how for each of the counties, it was roughly in 2020 when the boom of EV sales had occurred. This can be attributed to the pandemic and how the digital world was hyperfocused on as everyone was online during this time. Culture and status were heavily perceived, and once the technology grew in popularity, so did use.

Ultimately, my story can be conveyed through each of these graphs. The bar chart depicts how in the beginning of my story, I wanted to see where EV car sales were the most concentrated and how many there were relative to more rural areas. From there, I wanted to see the data geographically, how EV cars were spreading as time increased and if EV's were staying close to high status and income areas or if they were becoming more popular in general. And lastly, the bar chart allows me to see the growth for each country, describing how it was in the last 5 years that EV's really took off, supporting my hypothesis that it is the high-income and status oriented counties that contain the most EV sales.